

1 Simulation 1: inference for the single pair comparison

We will try different distribution $\mathcal{X}$, function $\theta_i^*(\mathbf{x})$ with $i = 1, \dots, n$, some region Ω of $\mathbf{X}$, during the simulation, and thus we will specify them later. For $\mathbf{A}$, we define

$$|\mathbf{A}| = \#\{(i, j) \mid n \geq i > j \geq 1, A_{ij} = 1\}. \quad (1)$$

Consider something like $\mathcal{X} = \text{Unif}(0, 1)$ and $\theta_i^*(\mathbf{x}) = \mathbf{x} \sin(i \cdot \pi/8)$, and $i_0 = 1, j_0 = 4$. $S = 3$. $\Omega = (0.3, 0.8)$.

Some other notations: $\psi'(t) = \exp(t)/\{\exp(t) + 1\}^2$. $\delta(i \mid \mathbf{A}) = \{j \mid j \in [n], A_{ij} = 1\}$ a.k.a., all items j that have been compared with item i .

Algorithm 1: The generation process of $\mathcal{D}_n$

Input: Sampling probability p , Number of items n , the number of comparisons L .

Output: $\mathcal{D}_n = \{(\mathbf{X}_{ijl}, Y_{ijl}) \mid \text{when } A_{ij} = 1, i > j, i \in [n] \text{ and } l = 1, 2, \dots, L\}$

- Generating Erdos-Renyi graph:
 - Initialize $\mathbf{A} = [A_{ij}]_{(i,j) \in [n]^2}$ as a $n \times n$ matrix with all entries = 0.
 - For all $(i, j) \in \{n \geq i > j \geq 1\}$, we generate $A_{ij} = 1$ with probability p and $A_{ij} = 0$ with probability $1 - p$ (Bernoulli distribution).
 - For all $(i, j) \in \{n \geq i > j \geq 1\}$, set $A_{ji} = A_{ij}$.
- Generating $(\mathbf{X}_{ijl})$: For all $(i, j) \in \{n \geq i > j \geq 1\}$, $A_{ij} = 1$ and $l = 1, \dots, L = 10^4$ or 10^3 , we iid generate $X_{ijl} \sim \mathcal{X}$.
- Generating (Y_{ijl}) : For all $(i, j) \in \{n \geq i > j \geq 1\}$, $A_{ij} = 1$ and $l = 1, \dots, L = 10^4$ or 10^3 , we iid generate Y_{ijl} such that

$$Y_{ijl} = 1 \text{ with probability } \frac{\exp\{\theta_i^*(\mathbf{X}_{ijl})\}}{\exp\{\theta_i^*(\mathbf{X}_{ijl})\} + \exp\{\theta_j^*(\mathbf{X}_{ijl})\}}, \quad (2)$$

$$\text{and } Y_{ijl} = 0 \text{ with probability } 1 - \frac{\exp\{\theta_i^*(\mathbf{X}_{ijl})\}}{\exp\{\theta_i^*(\mathbf{X}_{ijl})\} + \exp\{\theta_j^*(\mathbf{X}_{ijl})\}}.$$

Algorithm 2: Estimating $\hat{\theta}_i(\cdot)$ for $i = 1, \dots, n$

Input: Data set $\mathcal{D}_n$.

Output: $\hat{\theta}_i(\mathbf{x})$ for $i = 1, \dots, n$

- Let $\mathcal{F}_\beta$ be the class of ReLU functions parametrized appropriately (we can try different parameters β to see the performance).
- Run SGD to minimize:

$$\begin{aligned}
& - \sum_{(i,j): i > j \text{ and } A_{ij}=1} \left[\sum_{l=1}^{L(\mathcal{D}_n)} Y_{ijl} \log \left[\psi \{ \theta_i(\mathbf{X}_{ijl} \mid \beta_i) - \theta_j(\mathbf{X}_{ijl} \mid \beta_j) \} \right] \right. \\
& \quad \left. + (1 - Y_{ijl}) \log \left[1 - \psi \{ \theta_i(\mathbf{X}_{ijl} \mid \beta_i) - \theta_j(\mathbf{X}_{ijl} \mid \beta_j) \} \right] \right], \tag{3}
\end{aligned}$$

over all possible $\theta_1, \dots, \theta_n \in \mathcal{F}_\beta$. The functions $\theta_1, \dots, \theta_n \in \mathcal{F}_\beta$ that SGD converges to, are our estimators $\hat{\theta}_1, \dots, \hat{\theta}_n$, respectively.

- if ReLU is still hard, consider function class like $\{\theta(\mathbf{x}) = \beta\mathbf{x} + \gamma\}$.
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Algorithm 3: Estimating $\hat{\pi}_i(\mathbf{x})$ for $i = 1, \dots, n$

Input: $\mathbf{A}$ and $\hat{\theta}_i(\mathbf{x})$ for $i = 1, \dots, n$

Output: $\hat{\pi}_i(\mathbf{x})$ for $i = 1, \dots, n$

- Obtain matrix $\hat{\Xi}(\mathbf{x}) = \left[A_{ij} \psi' \left\{ \hat{\theta}_i(\mathbf{x}) - \hat{\theta}_j(\mathbf{x}) \right\} \right]_{(i,j) \in [n]^2}$ and A_{ij} is the ij th entry of $\mathbf{A}$.
- Obtain matrix $\hat{\mathcal{L}}(\mathbf{x}) = \text{diag} \left\{ \hat{\Xi}(\mathbf{x}) \underbrace{(1, \dots, 1)^\top}_{n \text{ times}} \right\} - \hat{\Xi}(\mathbf{x})$.
- Obtain the $\hat{\mathcal{L}}^\dagger(\mathbf{x})$, the Moore-Penrose inverse of $\hat{\mathcal{L}}(\mathbf{x})$.
- Obtain $(\hat{\pi}_1(\mathbf{x}), \dots, \hat{\pi}_n(\mathbf{x}))^\top = \mathbb{I}(\mathbf{x} \in \Omega) |\mathbf{A}| \hat{\mathcal{L}}^\dagger(\mathbf{x} \mid \mathbf{A}) (\mathbf{e}_{i_0} - \mathbf{e}_{j_0})$. $\mathbf{e}_i$ is the standard basis of $\mathbb{R}^n$.

$$\text{Get } \hat{\mathcal{L}}^\dagger(\mathbf{x} \mid \mathbf{A}) (\mathbf{e}_{i_0} - \mathbf{e}_{j_0}) \Leftrightarrow \text{Get } \mathbf{b} \text{ such that } \begin{pmatrix} \hat{\mathcal{L}}(\mathbf{x} \mid \mathbf{A}) \\ \mathbf{1}_n^\top \end{pmatrix} \mathbf{b} = \begin{pmatrix} \mathbf{e}_{i_0} - \mathbf{e}_{j_0} \\ \mathbf{0}_n^\top \end{pmatrix} \tag{4}$$

Algorithm 4: The principal debiased estimator for $\mathcal{Q}_{i_0j_0}(\Omega)$ and its variance estimation

Input: Comparison graph $\mathbf{A}$, samples $\mathcal{D}_n^*$, number of cross-fittings S and domain Ω .

Output: $\hat{\mathcal{Q}}_{i_0j_0}(\Omega) = \sum_{s=1}^S \hat{\mathcal{Q}}_{i_0j_0}^{(s)}(\Omega)/S$ and its variance estimator $\hat{V}_{i_0j_0}(\Omega) = \sum_{s=1}^S \hat{V}_{i_0j_0}^{(s)}(\Omega)/S$.

- Split $[L] = \{1, \dots, L\}$ equally into S subsets, namely, $\mathcal{J}_n^{(1)}, \dots, \mathcal{J}_n^{(S)}$, such that $\mathcal{J}_n^{(1)} \cup \dots \cup \mathcal{J}_n^{(S)} = [L]$. We then split $\mathcal{D}_n^*$ into $\mathcal{D}_n^{(1)}, \dots, \mathcal{D}_n^{(S)}$ as $\mathcal{D}_n^{(s)} = \{(\mathbf{X}_{ijl}, Y_{ijl}) \mid n \geq i > j \geq 1, A_{ij} = 1, l \in \mathcal{J}_n^{(s)}\}$;
- For $s = 1, \dots, S$, input $\mathcal{D}_n^{(-s)} = \mathcal{D}_n^* \setminus \mathcal{D}_n^{(s)}$ as $\mathcal{D}_n$ in Algorithm 2 and the outputs of Algorithm 2 are denoted by $\hat{\theta}_1^{(s)}(\mathbf{x}), \dots, \hat{\theta}_n^{(s)}(\mathbf{x})$;
- For $s = 1, \dots, S$ and all (i, j) such that $A_{ij} = 1$ and $n \geq i > j \geq 1$ and all $l \in \mathcal{J}_n^{(s)}$, obtain

$$(\hat{\pi}_1^{(s)}(\mathbf{X}_{ijl}), \dots, \hat{\pi}_n^{(s)}(\mathbf{X}_{ijl}))$$

as the output of Algorithm 3 with $\mathbf{A} = \mathbf{A}$ and $(\hat{\theta}_1(\mathbf{x}), \dots, \hat{\theta}_n(\mathbf{x})) = (\hat{\theta}_1^{(s)}(\mathbf{X}_{ijl}), \dots, \hat{\theta}_n^{(s)}(\mathbf{X}_{ijl}))$.

- For each $s \in [S]$, obtain the test statistics $\hat{\mathcal{Q}}_{i_0j_0}^{(s)}(\Omega)$ as

$$\begin{aligned} \frac{1}{|\mathbf{A}|L/S} \sum_{(\mathbf{X}_{ijl}, y_{ijl}) \in \mathcal{D}_n^{(s)}} & \left[\mathbb{I}(\mathbf{X}_{ijl} \in \Omega) \left\{ \hat{\theta}_{i_0}^{(s)}(\mathbf{X}_{ijl}) - \hat{\theta}_{j_0}^{(s)}(\mathbf{X}_{ijl}) \right\} \right. \\ & \left. + \left\{ \hat{\pi}_i^{(s)}(\mathbf{X}_{ijl}) - \hat{\pi}_j^{(s)}(\mathbf{X}_{ijl}) \right\} \left[y_{ijl} - \psi \left\{ \hat{\theta}_i^{(s)}(\mathbf{X}_{ijl}) - \hat{\theta}_j^{(s)}(\mathbf{X}_{ijl}) \right\} \right] \right]. \end{aligned}$$

- For each $s \in [S]$, obtain the variance estimator $\hat{V}_{i_0j_0}^{(s)}(\Omega)$ as

$$\begin{aligned} \hat{\sigma}^{(s)}(\mathbf{A}) = \frac{1}{|\mathbf{A}|L/S} \sum_{i>j, A_{ij}=1, l \in \mathcal{J}_n^{(s)}} & \left[\mathbb{I}(\mathbf{X}_{ijl} \in \Omega) \left(\frac{1}{\sum_{j' \in \delta(i_0|\mathbf{A})} \psi' \{ \hat{\theta}_{i_0}^{(s)}(\mathbf{X}_{ijl}) - \hat{\theta}_{j'}^{(s)}(\mathbf{X}_{ijl}) \}} \right. \right. \\ & \left. \left. + \frac{1}{\sum_{i' \in \delta(j_0|\mathbf{A})} \psi' \{ \hat{\theta}_{j_0}^{(s)}(\mathbf{X}_{ijl}) - \hat{\theta}_{i'}^{(s)}(\mathbf{X}_{ijl}) \}} \right) \right]. \end{aligned}$$

$$\hat{V}_{i_0j_0}^{(s)}(\Omega) = \frac{1}{|\mathbf{A}|^2 L^2 / S^2} \sum_{i>j, A_{ij}=1, l \in \mathcal{J}_n^{(s)}} \left\{ \mathbb{I}(\mathbf{X}_{ijl} \in \Omega) \{ \hat{\theta}_{i_0}^{(s)}(\mathbf{X}_{ijl}) - \hat{\theta}_{j_0}^{(s)}(\mathbf{X}_{ijl}) \} - \hat{\mathcal{Q}}_{i_0j_0}(\Omega) \right\}^2 + \frac{1}{L} \hat{\sigma}^{(s)}(\mathbf{A}). \quad (5)$$

Algorithm 5: Simulation implementation

- We will specify all parameters in each specific numerical experiment. We will try many different parameters and the record the corresponding results.
- Calculate

$$\mathcal{Q}_{i_0 j_0}(\Omega) = \frac{1}{10^5} \sum_{i=1, \dots, 10^5} \mathbb{I}(\mathbf{X}_i \in \Omega) \cdot \{\theta_{i_0}^*(\mathbf{X}_i) - \theta_{j_0}^*(\mathbf{X}_i)\}, \quad (6)$$

where $\mathbf{X}_i$ are iid generated from $\mathcal{X}$.

- 1 Obtain $\mathcal{D}_n^*$ from Algorithm 1.
- 2 Obtain $\hat{\mathcal{Q}}_{i_0 j_0}(\Omega)$ and $\hat{V}_{i_0 j_0}(\Omega)$ from Algorithm 4.
- 3 Obtain the 95% CI

$$\left(\hat{\mathcal{Q}}_{i_0 j_0}(\Omega) - \sqrt{\hat{V}_{i_0 j_0}(\Omega)} \cdot z_{.975}, \quad \hat{\mathcal{Q}}_{i_0 j_0}(\Omega) + \sqrt{\hat{V}_{i_0 j_0}(\Omega)} \cdot z_{.975} \right), \quad (7)$$

where $z_{.975} = 1.96$ is the 0.975 quantile of the standard Gaussian distribution.

- Repeat the above procedures 1–3, $T = 100$ times
- (Empirical coverage rate) Among the T times, record the empirical coverage rate as

$$\frac{\text{Number of times that the CI contains } \mathcal{Q}_{i_0 j_0}(\Omega)}{T}. \quad (8)$$

- (Estimation error) Among the T times, record $\hat{\mathcal{Q}}_{i_0 j_0}(\Omega) - \mathcal{Q}_{i_0 j_0}(\Omega)$ in all rounds, and we will use a violin plot to present the estimation error.
- (Average CI length) Among the T times, record $2\sqrt{\hat{V}_{i_0 j_0}(\Omega)} \cdot z_{.975}$ in all rounds, and we will use the average of them to show the average confidence interval length.
- (Estimation Error of the trivial estimator) In each round T , calculate the estimation error of the trivial estimator:

$$\begin{aligned} \hat{\mathcal{Q}}_{i_0 j_0}^{(s), \text{trivial}}(\Omega) &= \frac{1}{|\mathbf{A}|L/S} \sum_{(\mathbf{X}_{ijl}, y_{ijl}) \in \mathcal{D}_n^{(s)}} \left[\mathbb{I}(\mathbf{X}_{ijl} \in \Omega) \left\{ \hat{\theta}_{i_0}^{(s)}(\mathbf{X}_{ijl}) - \hat{\theta}_{j_0}^{(s)}(\mathbf{X}_{ijl}) \right\} \right] \\ \hat{\mathcal{Q}}_{i_0 j_0}^{\text{trivial}}(\Omega) &= S^{-1} \sum_{s=1}^S \hat{\mathcal{Q}}_{i_0 j_0}^{(s), \text{trivial}}(\Omega). \end{aligned} \quad (9)$$

We report the trivial estimator error $\hat{\mathcal{Q}}_{i_0 j_0}^{\text{trivial}}(\Omega) - \mathcal{Q}_{i_0 j_0}(\Omega)$.
